04 Enrichment analysis

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Setup

This script keeps only the **GSEA** part of downstream enrichment analysis.
GSEA is performed on the full ranked gene list using **GO BP**, **KEGG**, and **Hallmark** gene sets.

```
if (!exists("project_root")) {
  cur <- tryCatch(knitr::current_input(), error = function(e) NULL)
  path <- if (is.null(cur)) NULL else if (is.character(cur)) cur else cur[["path"]]
  if (!is.null(path) && length(path) > 0) {
    project_root <- normalizePath(file.path(dirname(path[1]), ".."), mustWork = FALSE)
  } else {
    wd <- getwd()
    project_root <- if (dir.exists(file.path(wd, "Data"))) wd else normalizePath(file.path(wd, ".."), mu
  }
}
INPUT_DIR       <- file.path(project_root, "Results", "02_DEG")
RESULTS_DIR     <- file.path(project_root, "Results", "04_Enrichment")
TUMOR_VS_NORMAL <- file.path(INPUT_DIR, "Tumor_vs_normal.csv")
```

Packages

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:dbplyr':
##
##     ident, sql

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tibble)
library(ggplot2)
library(clusterProfiler)
```

```
##

## Registered S3 methods overwritten by 'treeio':
##   method               from
##   MRCA.phylo           tidytree
##   MRCA.treedata        tidytree
##   Nnode.treedata       tidytree
##   Ntip.treedata        tidytree
##   ancestor.phylo       tidytree
##   ancestor.treedata    tidytree
##   child.phylo          tidytree
##   child.treedata       tidytree
##   full_join.phylo      tidytree
##   full_join.treedata   tidytree
##   groupClade.phylo     tidytree
##   groupClade.treedata  tidytree
##   groupOTU.phylo       tidytree
##   groupOTU.treedata    tidytree
##   is.rooted.treedata   tidytree
##   nodeid.phylo         tidytree
##   nodeid.treedata      tidytree
##   nodelab.phylo        tidytree
##   nodelab.treedata     tidytree
##   offspring.phylo      tidytree
##   offspring.treedata   tidytree
##   parent.phylo         tidytree
##   parent.treedata      tidytree
##   root.treedata        tidytree
##   rootnode.phylo       tidytree
##   sibling.phylo        tidytree

## clusterProfiler v4.6.2  For help: https://yulab-smu.top/biomedical-knowledge-mining-book/
##
## If you use clusterProfiler in published research, please cite:
## T Wu, E Hu, S Xu, M Chen, P Guo, Z Dai, T Feng, L Zhou, W Tang, L Zhan, X Fu, S Liu, X Bo, and G Yu.

##
## Attaching package: 'clusterProfiler'
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```r
library(org.Hs.eg.db)
```

```
## Loading required package: AnnotationDbi

## Loading required package: stats4

## Loading required package: BiocGenerics

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:dplyr':
##
##     combine, intersect, setdiff, union

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename,
##     cbind, colnames, dirname, do.call, duplicated, eval,
##     evalq, Filter, Find, get, grep, grepl, intersect,
##     is.unsorted, lapply, Map, mapply, match, mget, order,
##     paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, setdiff, sort, table,
##     tapply, union, unique, unsplit, which.max, which.min

## Loading required package: Biobase

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages
##     'citation("pkgname")'.

## Loading required package: IRanges

## Loading required package: S4Vectors

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:clusterProfiler':
##
##     rename

## The following objects are masked from 'package:dplyr':
##
##     first, rename

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
```

```
##
## Attaching package: 'IRanges'

## The following object is masked from 'package:clusterProfiler':
##
##     slice

## The following objects are masked from 'package:dplyr':
##
##     collapse, desc, slice

##
## Attaching package: 'AnnotationDbi'

## The following object is masked from 'package:clusterProfiler':
##
##     select

## The following object is masked from 'package:dplyr':
##
##     select

##
```

```r
library(enrichplot)
library(msigdbr)
```

Read differential expression results

```r
res <- read.csv(TUMOR_VS_NORMAL, row.names = 1)
stopifnot("logFC" %in% names(res), "adj.P.Val" %in% names(res))

res$gene_id <- sub("\\..*$", "", rownames(res))
message("Genes in ranked list input: ", nrow(res))
```

```
## Genes in ranked list input: 17075
```

Map ENSEMBL to ENTREZID

```r
id_map <- AnnotationDbi::select(
  org.Hs.eg.db,
  keys = unique(res$gene_id),
  keytype = "ENSEMBL",
  columns = c("ENTREZID", "SYMBOL")
)
```

```
## 'select()' returned 1:many mapping between keys and columns
```

```r
id_map <- id_map %>%
  filter(!is.na(ENTREZID)) %>%
  distinct(ENSEMBL, .keep_all = TRUE)
```

Ranked gene list for GSEA

```
rank_df <- res %>%
  left_join(id_map, by = c("gene_id" = "ENSEMBL")) %>%
  filter(!is.na(ENTREZID)) %>%
  group_by(ENTREZID) %>%
  slice_max(order_by = abs(logFC), n = 1, with_ties = FALSE) %>%
  ungroup()

gene_list <- rank_df$logFC
names(gene_list) <- rank_df$ENTREZID
gene_list <- sort(gene_list, decreasing = TRUE)

head(gene_list)
```

```
##     1300     1469     4322     3381     4320     1301
## 6.861573 6.586908 6.332983 5.849066 5.789660 5.329859
```

GO GSEA

```
gse_go <- gseGO(
  geneList = gene_list,
  OrgDb = org.Hs.eg.db,
  keyType = "ENTREZID",
  ont = "BP",
  pAdjustMethod = "BH",
  minGSSize = 10,
  maxGSSize = 500,
  verbose = FALSE
)
```

```
## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : There were 9 pathways for which P-values were
## not calculated properly due to unbalanced (positive and negative)
## gene-level statistic values. For such pathways pval, padj, NES,
## log2err are set to NA. You can try to increase the value of the
## argument nPermSimple (for example set it nPermSimple = 10000)

## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : For some of the pathways the P-values were
## likely overestimated. For such pathways log2err is set to NA.

## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : For some pathways, in reality P-values are less
## than 1e-10. You can set the `eps` argument to zero for better
## estimation.
```

```
gse_go_table <- as.data.frame(gse_go) %>%
  arrange(p.adjust)

gse_go_table %>% slice_head(n = 10)
```

```
##                     ID
## GO:0000070 GO:0000070
## GO:0065004 GO:0065004
```

```
## GO:0006334 GO:0006334
## GO:0000819 GO:0000819
## GO:0071459 GO:0071459
## GO:0071824 GO:0071824
## GO:0008608 GO:0008608
## GO:1905818 GO:1905818
## GO:0033045 GO:0033045
## GO:0034728 GO:0034728
##                                                    Description
## GO:0000070                   mitotic sister chromatid segregation
## GO:0065004                           protein-DNA complex assembly
## GO:0006334                                    nucleosome assembly
## GO:0000819                           sister chromatid segregation
## GO:0071459 protein localization to chromosome, centromeric region
## GO:0071824               protein-DNA complex subunit organization
## GO:0008608      attachment of spindle microtubules to kinetochore
## GO:1905818                     regulation of chromosome separation
## GO:0033045             regulation of sister chromatid segregation
## GO:0034728                                 nucleosome organization
##            setSize enrichmentScore      NES pvalue     p.adjust
## GO:0000070     197       0.6327727 2.869905  1e-10 1.023115e-08
## GO:0065004     195       0.6307310 2.856091  1e-10 1.023115e-08
## GO:0006334      86       0.7121264 2.798095  1e-10 1.023115e-08
## GO:0000819     223       0.6148365 2.787184  1e-10 1.023115e-08
## GO:0071459      38       0.8291249 2.754098  1e-10 1.023115e-08
## GO:0071824     229       0.5979225 2.747230  1e-10 1.023115e-08
## GO:0008608      43       0.7830672 2.728720  1e-10 1.023115e-08
## GO:1905818     105       0.6691607 2.689884  1e-10 1.023115e-08
## GO:0033045     101       0.6673999 2.674106  1e-10 1.023115e-08
## GO:0034728     120       0.6592674 2.665120  1e-10 1.023115e-08
##                  qvalue rank                    leading_edge
## GO:0000070 7.572045e-09 1748 tags=31%, list=10%, signal=29%
## GO:0065004 7.572045e-09 2005 tags=36%, list=12%, signal=32%
## GO:0006334 7.572045e-09 1039  tags=47%, list=6%, signal=44%
## GO:0000819 7.572045e-09 1748 tags=30%, list=10%, signal=27%
## GO:0071459 7.572045e-09 2318 tags=71%, list=14%, signal=61%
## GO:0071824 7.572045e-09 2045 tags=33%, list=12%, signal=29%
## GO:0008608 7.572045e-09  595  tags=40%, list=4%, signal=38%
## GO:1905818 7.572045e-09 2408 tags=42%, list=14%, signal=36%
## GO:0033045 7.572045e-09  491  tags=28%, list=3%, signal=27%
## GO:0034728 7.572045e-09 2143 tags=44%, list=13%, signal=39%
##
## GO:0000070                                                                  4751/11065/24137/332/146909/
## GO:0065004                             55355/8357/8348/9787/8343/1063/8356/55723/1058/8360/8318/8358/8
## GO:0006334
## GO:0000819                                          4751/11065/24137/332/146909/83540/7153/9928/9787/99
## GO:0071459
## GO:0071824 55355/8357/8348/9787/8343/1063/8356/55723/1058/8360/8318/8358/8367/8344/3017/8345/81620/83
## GO:0008608
## GO:1905818
## GO:0033045
## GO:0034728
```

KEGG GSEA

```r
gse_kegg <- gseKEGG(
  geneList = gene_list,
  organism = "hsa",
  pAdjustMethod = "BH",
  minGSSize = 10,
  maxGSSize = 500,
  verbose = FALSE
)
```

```
## Reading KEGG annotation online: "https://rest.kegg.jp/link/hsa/pathway"...
```

```
## Reading KEGG annotation online: "https://rest.kegg.jp/list/pathway/hsa"...
```

```
## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : For some of the pathways the P-values were
## likely overestimated. For such pathways log2err is set to NA.
```

```
## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : For some pathways, in reality P-values are less
## than 1e-10. You can set the `eps` argument to zero for better
## estimation.
```

```r
if (!is.null(gse_kegg) && nrow(as.data.frame(gse_kegg)) > 0) {
  gse_kegg <- setReadable(gse_kegg, OrgDb = org.Hs.eg.db, keyType = "ENTREZID")
}

gse_kegg_table <- as.data.frame(gse_kegg) %>%
  arrange(p.adjust)

gse_kegg_table %>% slice_head(n = 10)
```

```
##                ID                            Description setSize
## hsa04110 hsa04110                             Cell cycle     156
## hsa05322 hsa05322            Systemic lupus erythematosus     104
## hsa04613 hsa04613 Neutrophil extracellular trap formation     161
## hsa05203 hsa05203                     Viral carcinogenesis     185
## hsa04519 hsa04519                       Cadherin signaling     375
## hsa05034 hsa05034                              Alcoholism     156
## hsa04923 hsa04923   Regulation of lipolysis in adipocytes      55
## hsa00982 hsa00982       Drug metabolism - cytochrome P450      53
## hsa04020 hsa04020               Calcium signaling pathway     234
## hsa03030 hsa03030                          DNA replication      35
##          enrichmentScore       NES       pvalue     p.adjust
## hsa04110       0.5900556  2.600294 1.000000e-10 8.825000e-09
## hsa05322       0.6255796  2.578844 1.000000e-10 8.825000e-09
## hsa04613       0.5080953  2.232723 1.000000e-10 8.825000e-09
## hsa05203       0.4765772  2.147652 1.000000e-10 8.825000e-09
## hsa04519      -0.4777043 -1.700351 4.138620e-09 2.921866e-07
## hsa05034       0.4737429  2.087720 6.470453e-09 3.806783e-07
## hsa04923      -0.7044362 -2.126941 2.903527e-08 1.464207e-06
## hsa00982      -0.6996430 -2.100764 1.114341e-07 4.917028e-06
## hsa04020      -0.4949715 -1.722342 2.350949e-07 9.220945e-06
## hsa03030       0.7069309  2.389868 2.726827e-07 9.291776e-06
##                qvalue rank                    leading_edge
```

```
## hsa04110 6.026316e-09 2408 tags=36%, list=14%, signal=31%
## hsa05322 6.026316e-09 1386  tags=47%, list=8%, signal=44%
## hsa04613 6.026316e-09 1968 tags=34%, list=12%, signal=30%
## hsa05203 6.026316e-09 3070 tags=35%, list=18%, signal=29%
## hsa04519 1.995250e-07 3589 tags=39%, list=21%, signal=32%
## hsa05034 2.599533e-07 1968 tags=38%, list=12%, signal=34%
## hsa04923 9.998613e-07 2484 tags=51%, list=15%, signal=44%
## hsa00982 3.357684e-06 3088 tags=55%, list=18%, signal=45%
## hsa04020 6.296694e-06 4043 tags=45%, list=24%, signal=35%
## hsa03030 6.345062e-06 3584 tags=71%, list=21%, signal=56%
##
## hsa04110
## hsa05322
## hsa04613
## hsa05203
## hsa04519 MYO7A/RRAS/KRT32/LRP6/PCDH8/PCDHB13/VCL/CDH26/ANKRD6/LRP5/PCDHB14/LIMA1/GABRR1/WWTR1/PCDHAC
## hsa05034
## hsa04923
## hsa00982
## hsa04020
## hsa03030
```

Hallmark GSEA

```r
hallmark_raw <- msigdbr(species = "Homo sapiens", collection = "H")
hallmark_raw$entrez_gene <- hallmark_raw$ncbi_gene

hallmark <- hallmark_raw %>%
  dplyr::select(gs_name, entrez_gene)

gsea_hallmark <- GSEA(
  geneList = gene_list,
  TERM2GENE = hallmark,
  pAdjustMethod = "BH",
  minGSSize = 10,
  maxGSSize = 500,
  verbose = FALSE
)
```

```
## Warning in fgseaMultilevel(pathways = pathways, stats = stats,
## minSize = minSize, : For some pathways, in reality P-values are less
## than 1e-10. You can set the `eps` argument to zero for better
## estimation.
```

```r
gsea_hallmark_table <- as.data.frame(gsea_hallmark) %>%
  arrange(p.adjust)

gsea_hallmark_table %>% slice_head(n = 10)
```

```
##                                                                    ID
## HALLMARK_E2F_TARGETS                              HALLMARK_E2F_TARGETS
## HALLMARK_G2M_CHECKPOINT                         HALLMARK_G2M_CHECKPOINT
## HALLMARK_MYC_TARGETS_V1                         HALLMARK_MYC_TARGETS_V1
## HALLMARK_ADIPOGENESIS                             HALLMARK_ADIPOGENESIS
```

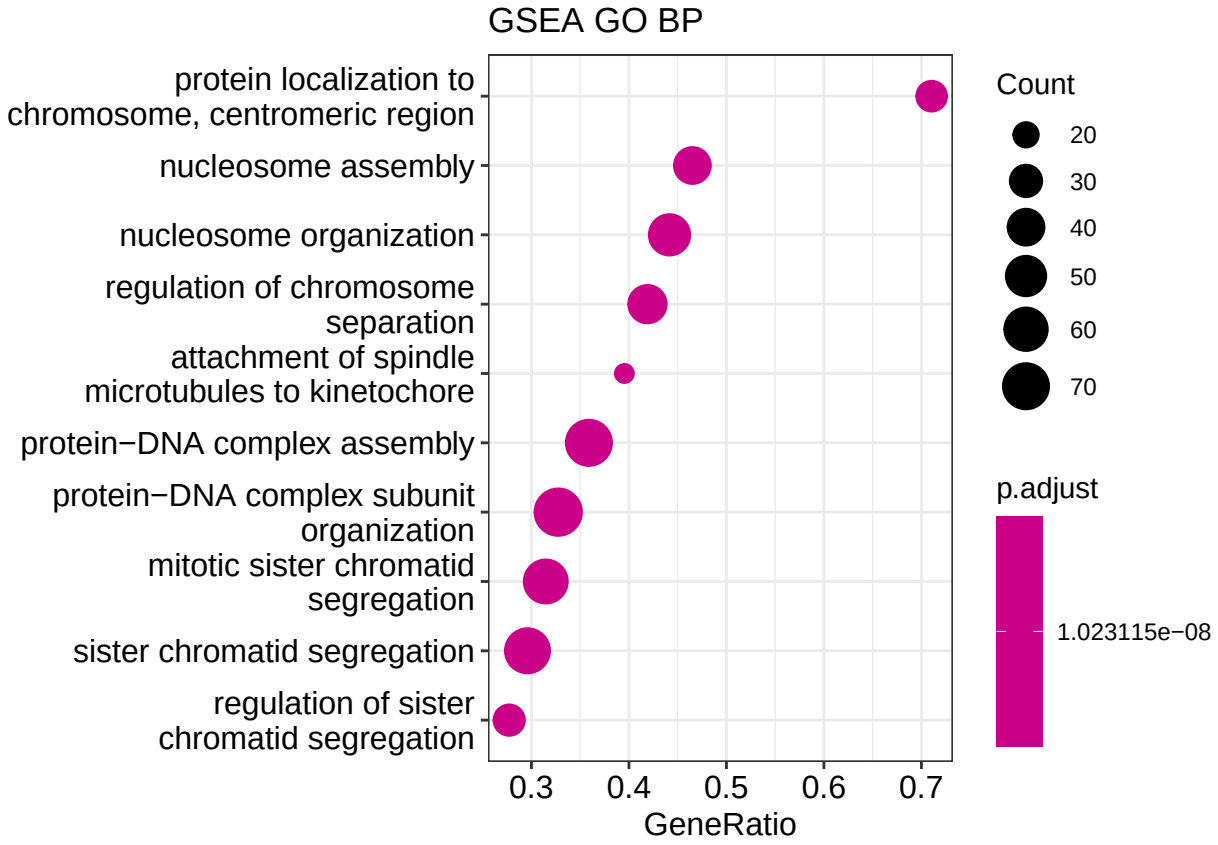
```
## HALLMARK_MITOTIC_SPINDLE                     HALLMARK_MITOTIC_SPINDLE
## HALLMARK_MTORC1_SIGNALING                   HALLMARK_MTORC1_SIGNALING
## HALLMARK_MYOGENESIS                               HALLMARK_MYOGENESIS
## HALLMARK_DNA_REPAIR                               HALLMARK_DNA_REPAIR
## HALLMARK_MYC_TARGETS_V2                       HALLMARK_MYC_TARGETS_V2
## HALLMARK_INTERFERON_ALPHA_RESPONSE HALLMARK_INTERFERON_ALPHA_RESPONSE
##                                                             Description
## HALLMARK_E2F_TARGETS                             HALLMARK_E2F_TARGETS
## HALLMARK_G2M_CHECKPOINT                       HALLMARK_G2M_CHECKPOINT
## HALLMARK_MYC_TARGETS_V1                       HALLMARK_MYC_TARGETS_V1
## HALLMARK_ADIPOGENESIS                           HALLMARK_ADIPOGENESIS
## HALLMARK_MITOTIC_SPINDLE                     HALLMARK_MITOTIC_SPINDLE
## HALLMARK_MTORC1_SIGNALING                   HALLMARK_MTORC1_SIGNALING
## HALLMARK_MYOGENESIS                               HALLMARK_MYOGENESIS
## HALLMARK_DNA_REPAIR                               HALLMARK_DNA_REPAIR
## HALLMARK_MYC_TARGETS_V2                       HALLMARK_MYC_TARGETS_V2
## HALLMARK_INTERFERON_ALPHA_RESPONSE HALLMARK_INTERFERON_ALPHA_RESPONSE
##                                   setSize enrichmentScore       NES
## HALLMARK_E2F_TARGETS                  200       0.7202345  3.235738
## HALLMARK_G2M_CHECKPOINT               199       0.6764064  3.036238
## HALLMARK_MYC_TARGETS_V1               200       0.5389512  2.421302
## HALLMARK_ADIPOGENESIS                 198      -0.5612480 -1.947722
## HALLMARK_MITOTIC_SPINDLE              198       0.4409634  1.979575
## HALLMARK_MTORC1_SIGNALING             200       0.4347607  1.953214
## HALLMARK_MYOGENESIS                   193      -0.5068398 -1.751620
## HALLMARK_DNA_REPAIR                   149       0.4219374  1.848325
## HALLMARK_MYC_TARGETS_V2                58       0.5745241  2.145937
## HALLMARK_INTERFERON_ALPHA_RESPONSE     97       0.4415498  1.822122
##                                        pvalue     p.adjust
## HALLMARK_E2F_TARGETS               1.000000e-10 1.666667e-09
## HALLMARK_G2M_CHECKPOINT            1.000000e-10 1.666667e-09
## HALLMARK_MYC_TARGETS_V1            1.000000e-10 1.666667e-09
## HALLMARK_ADIPOGENESIS              4.355610e-10 5.444513e-09
## HALLMARK_MITOTIC_SPINDLE           2.316055e-09 2.316055e-08
## HALLMARK_MTORC1_SIGNALING          8.098288e-09 6.748573e-08
## HALLMARK_MYOGENESIS                7.506898e-07 5.362070e-06
## HALLMARK_DNA_REPAIR                3.586079e-06 2.241299e-05
## HALLMARK_MYC_TARGETS_V2            4.325313e-06 2.402952e-05
## HALLMARK_INTERFERON_ALPHA_RESPONSE 6.953946e-05 3.476973e-04
##                                       qvalue rank
## HALLMARK_E2F_TARGETS               8.771930e-10 3024
## HALLMARK_G2M_CHECKPOINT            8.771930e-10 2437
## HALLMARK_MYC_TARGETS_V1            8.771930e-10 4910
## HALLMARK_ADIPOGENESIS              2.865533e-09 2958
## HALLMARK_MITOTIC_SPINDLE           1.218976e-08 1078
## HALLMARK_MTORC1_SIGNALING          3.551881e-08 4726
## HALLMARK_MYOGENESIS                2.822142e-06 3398
## HALLMARK_DNA_REPAIR                1.179631e-05 5328
## HALLMARK_MYC_TARGETS_V2            1.264711e-05 5377
## HALLMARK_INTERFERON_ALPHA_RESPONSE 1.829986e-04 4431
##                                                     leading_edge
## HALLMARK_E2F_TARGETS               tags=58%, list=18%, signal=48%
## HALLMARK_G2M_CHECKPOINT            tags=49%, list=14%, signal=42%
## HALLMARK_MYC_TARGETS_V1            tags=56%, list=29%, signal=40%
```

```
## HALLMARK_ADIPOGENESIS              tags=29%, list=17%, signal=24%
## HALLMARK_MITOTIC_SPINDLE            tags=16%, list=6%, signal=15%
## HALLMARK_MTORC1_SIGNALING          tags=54%, list=28%, signal=39%
## HALLMARK_MYOGENESIS                tags=41%, list=20%, signal=33%
## HALLMARK_DNA_REPAIR                tags=52%, list=31%, signal=36%
## HALLMARK_MYC_TARGETS_V2            tags=72%, list=32%, signal=50%
## HALLMARK_INTERFERON_ALPHA_RESPONSE tags=53%, list=26%, signal=39%
##
## HALLMARK_E2F_TARGETS               24137/332/146909/4605/7153/9833/6241/9787/991/79733/9212/4288/534
## HALLMARK_G2M_CHECKPOINT
## HALLMARK_MYC_TARGETS_V1                                                    991/8318/890/4085/3838/483
## HALLMARK_ADIPOGENESIS
## HALLMARK_MITOTIC_SPINDLE
## HALLMARK_MTORC1_SIGNALING                                                                          624
## HALLMARK_MYOGENESIS
## HALLMARK_DNA_REPAIR
## HALLMARK_MYC_TARGETS_V2
## HALLMARK_INTERFERON_ALPHA_RESPONSE
```

Plots

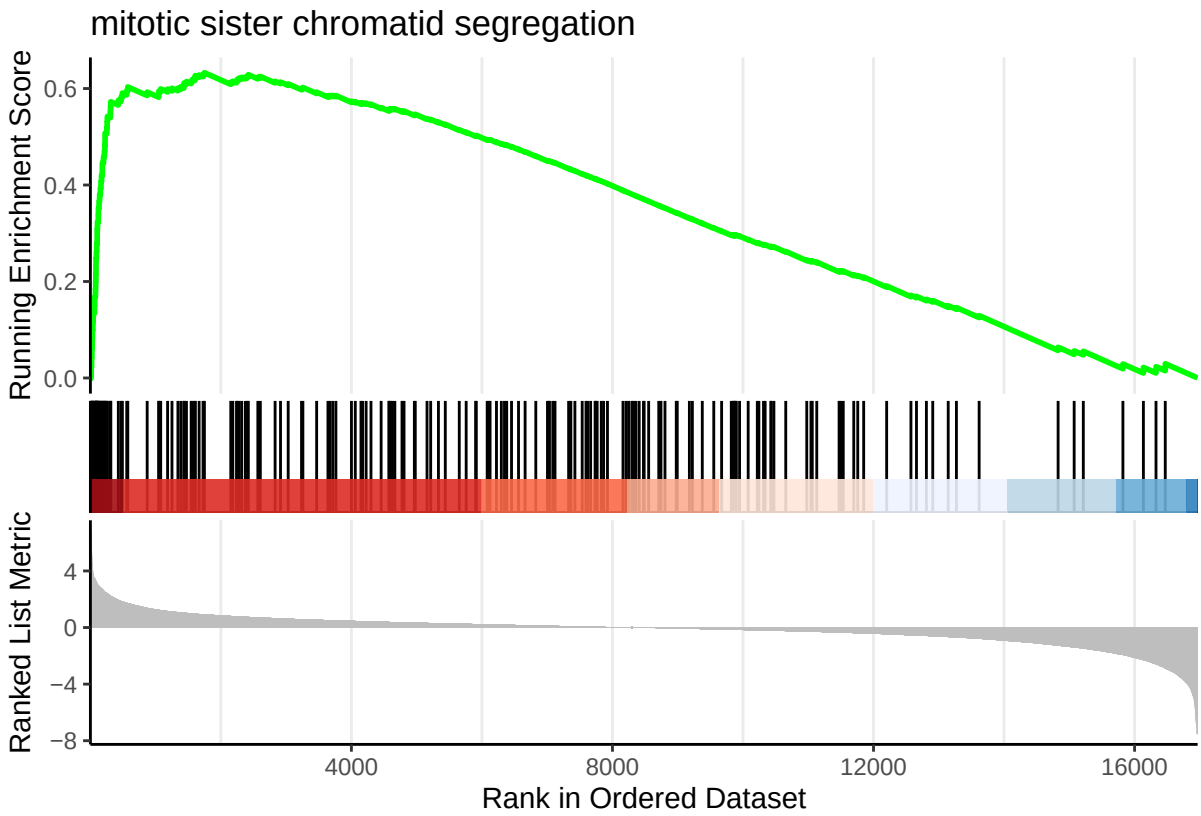
```r
dir.create(RESULTS_DIR, showWarnings = FALSE, recursive = TRUE)

p_gse_go <- dotplot(gse_go, showCategory = 10) + ggtitle("GSEA GO BP")
print(p_gse_go)
```



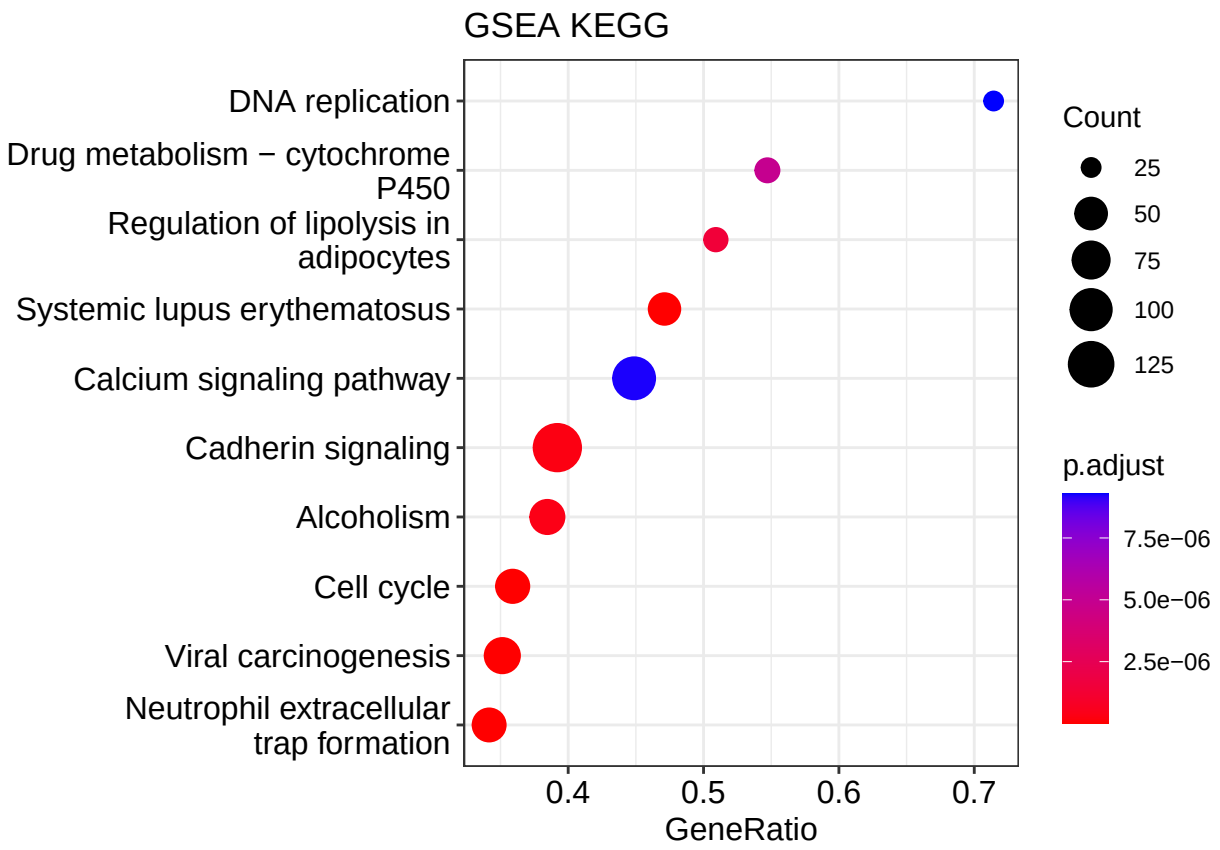
```r
ggsave(file.path(RESULTS_DIR, "GSEA_GO_dotplot.pdf"), p_gse_go, width = 9, height = 8)

top_id_go <- gse_go@result$ID[1]
top_title_go <- gse_go@result$Description[1]
p_go_one <- gseaplot2(gse_go, geneSetID = top_id_go, title = top_title_go)
print(p_go_one)
```



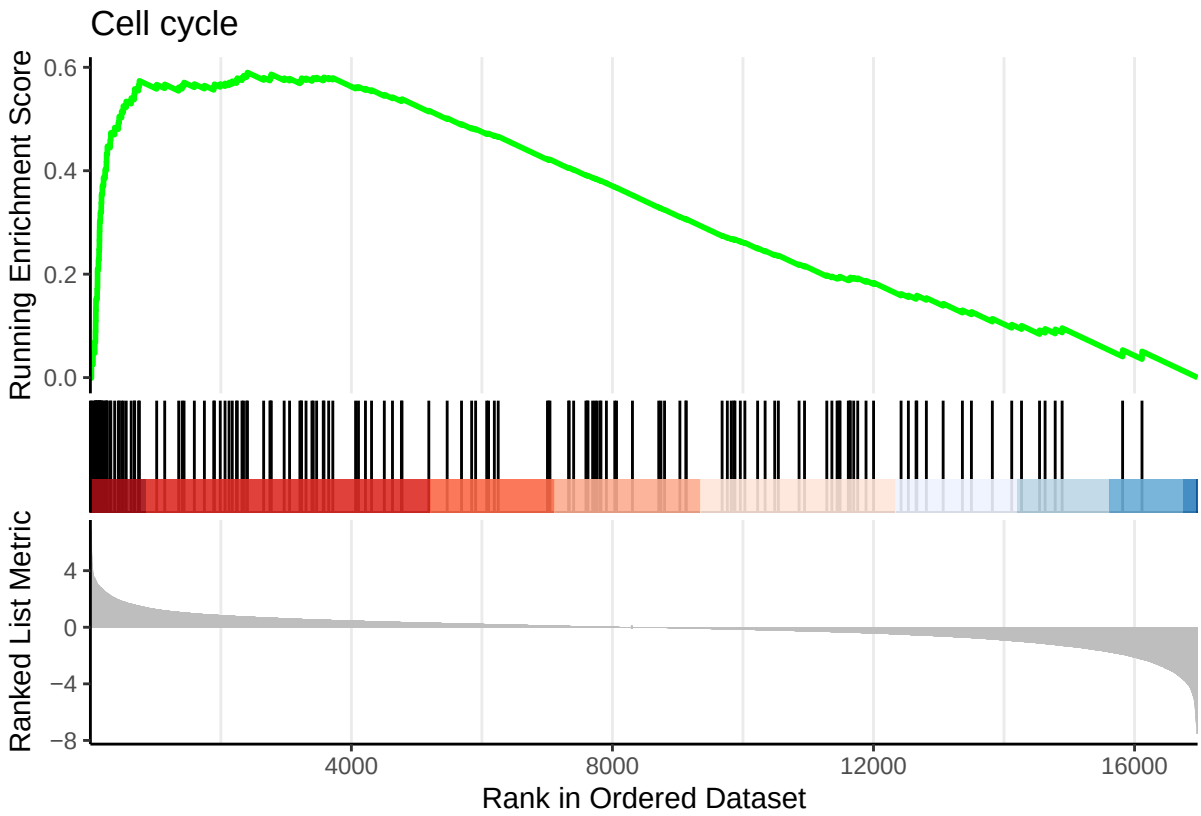

```r
ggsave(file.path(RESULTS_DIR, "GSEA_GO_1.pdf"), p_go_one, width = 9, height = 6.5)

p_gse_kegg <- dotplot(gse_kegg, showCategory = 10) + ggtitle("GSEA KEGG")
print(p_gse_kegg)
```



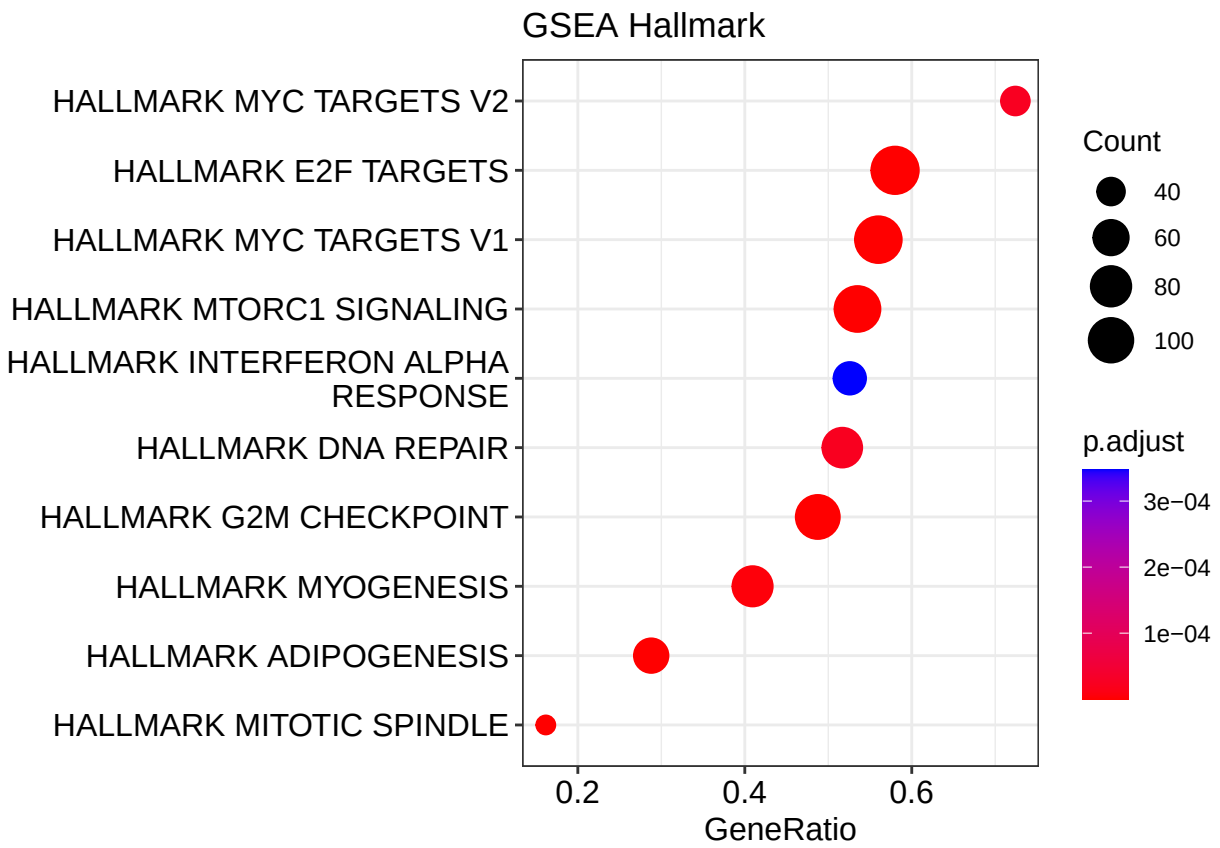

```r
ggsave(file.path(RESULTS_DIR, "GSEA_KEGG_dotplot.pdf"), p_gse_kegg, width = 9, height = 8)

top_id_kegg <- gse_kegg@result$ID[1]
top_title_kegg <- gse_kegg@result$Description[1]
p_kegg_one <- gseaplot2(gse_kegg, geneSetID = top_id_kegg, title = top_title_kegg)
print(p_kegg_one)
```



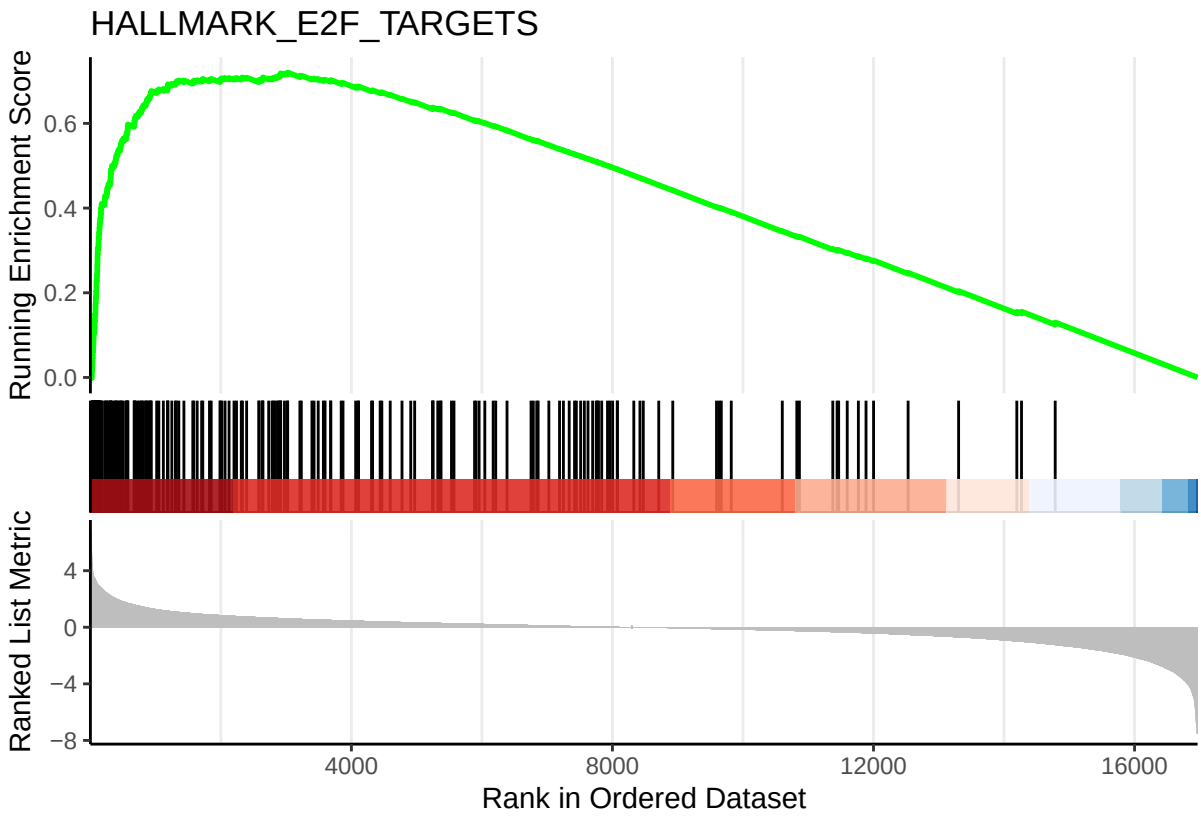

```
ggsave(file.path(RESULTS_DIR, "GSEA_KEGG_1.pdf"), p_kegg_one, width = 9, height = 6.5)

p_hallmark <- dotplot(gsea_hallmark, showCategory = 10) + ggtitle("GSEA Hallmark")
print(p_hallmark)
```




```
ggsave(file.path(RESULTS_DIR, "Hallmark_GSEA_dotplot.pdf"), p_hallmark, width = 9, height = 8)

top_id_hallmark <- gsea_hallmark@result$ID[1]
top_title_hallmark <- gsea_hallmark@result$Description[1]
p_hallmark_one <- gseaplot2(gsea_hallmark, geneSetID = top_id_hallmark, title = top_title_hallmark)
print(p_hallmark_one)
```




```
ggsave(file.path(RESULTS_DIR, "Hallmark_GSEA_1.pdf"), p_hallmark_one, width = 9, height = 6.5)
```

Save tables

```
write.csv(gse_go_table %>% slice_head(n = 10), file.path(RESULTS_DIR, "GSEA_GO.csv"), row.names = FALSE)
write.csv(gse_kegg_table %>% slice_head(n = 10), file.path(RESULTS_DIR, "GSEA_KEGG.csv"), row.names = F
write.csv(gsea_hallmark_table %>% slice_head(n = 10), file.path(RESULTS_DIR, "Hallmark_GSEA.csv"), row.

saveRDS(gse_go, file.path(RESULTS_DIR, "gse_go.rds"))
saveRDS(gse_kegg, file.path(RESULTS_DIR, "gse_kegg.rds"))
saveRDS(gsea_hallmark, file.path(RESULTS_DIR, "gsea_hallmark.rds"))
```